

CR-BIS Data Acquisition User Guide

Document Version: 1.0

Contents

- Introduction
- Part 1: Pre-calibration
 - 1.1 Measurement of Camera Delay Curve
 - 1.2 Determination of CR-BIS Total Exposure Time Formula
- Part 2: CR-BIS Data Acquisition Workflow
 - 2.1 Overview of CR-BIS Scripts and Toolbar
 - 2.2 Parameter Configuration
 - 2.3 Low-Magnification Map Acquisition
 - 2.4 Target Selection
 - 2.5 Production Data Acquisition
- Part 3: Preprocessing of CR-BIS Output Data
 - 3.1 Processing of TIF-format Data
 - 3.2 Processing of EER-format Data
 - 3.3 Processing of Cryo-ET Data

1. Introduction

This guide describes the use of the **CR-BIS (Continuous Recording Beam-Image Shift)** data acquisition script for **SerialEM**. Users can customize parameters as needed and perform multiple acquisition modes, including conventional BIS and CR-BIS. The following sections focus on the operation of the CR-BIS mode and its required pre-calibration steps.

Part 1: Pre-calibration

1.1 Measuring Camera Delay Curves

Different cryo-EM camera models exhibit distinct delay characteristics. It is recommended to measure the delay curve in advance to quantify the theoretical acceleration achievable with CR-BIS.

Use the script **Report_Clock_script.txt**. Set a series of exposure times and perform five repeated measurements for each setting. Plot the relationship between the set exposure time and actual recorded time to obtain the intercept and slope parameters.

```
start 0.3 s
3 frames were saved to X:\userdata\20250520_yq_test\K2_0006.tif
2.53 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0007.tif
2.53 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0008.tif
2.50 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0009.tif
2.63 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0010.tif
2.56 seconds elapsed time
start 0.6 s
6 frames were saved to X:\userdata\20250520_yq_test\K2_0011.tif
2.84 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0012.tif
2.83 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0013.tif
2.81 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0014.tif
2.83 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0015.tif
2.83 seconds elapsed time
start 1 s
10 frames were saved to X:\userdata\20250520_yq_test\K2_0016.tif
3.27 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0017.tif
3.25 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0018.tif
3.34 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0019.tif
3.23 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0020.tif
3.27 seconds elapsed time
```

SerialEM log output example

1.2 Determining the Formula for CR-BIS Total Exposure Time

In CR-BIS mode, the total set exposure time is defined as:

$$T_{total\ set} = (T_{set\ per\ point} + T_{blank}) * point\ number + T_{buffer} \quad (1)$$

Examples:

K3 camera: $T_{total\ set} = (T_{set\ per\ point} + 0.22\ s) * point\ number + 0.6\ s$

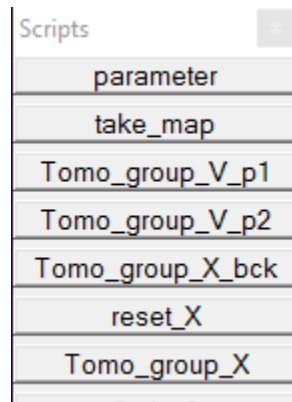
Falcon 4i camera: $T_{total\ set} = (T_{set\ per\ point} + 0.26\ s) * point\ number + 1.2\ s$

A CR-BIS acquisition collects multiple target points in a single continuous exposure. By analyzing frame rate and blank-frame data, one can estimate the beam blank/shift time and buffer duration.

Part 2: CR-BIS Data Acquisition Workflow

2.1 Overview of CR-BIS Scripts and Toolbar

The CR-BIS package includes several sub-scripts accessible through the SerialEM toolbar:



- **parameter** — Core parameter configuration script; must be executed before running other scripts.
- **take_map** — Captures low-magnification maps for target selection.
- **Tomo_group_V_p1/p2** — Performs precise alignment of target points on the low-magnification map (critical for cryo-ET).

- reset_X — Records key microscope parameters after setup and restores them before acquisition to ensure consistency.
- Tomo_group_X — Main data acquisition script for production runs.

2.2 Parameter Configuration

The parameters required for CR-BIS operation are shown in the figure below. Each colored section corresponds to a specific parameter group in the parameter script.

Script 1: parameter

```

MacroName parameter

#====Tomo_group_X settings=====
#---General tomo and SPA setting-----
start_angle = 10  # first image stage tilt angle in degree
exp_start_angle = 0 # expected first image stage tilt angle in degree
min_angle = -50   # minimum stage tilt angle of the whole data, > -66
max_angle = 70    # maximum stage tilt angle of the whole data < 70
step       = 3     # stage tilt step in degrees
min_defocus = -2.0 # minimum defocus value
max_defocus = -2.5 # maximum defocus value

#---microscope and camera setting-----
autoloader = 1  #set 1 for TEM with autoloader
              # 2 for not
flash_CFEG = 1  #set 1 for TEM with CFEG
              # 2 for not
camera = 1     #set 1 for using Falcon4/4i camera for data collection
              #2 for K3
              #3 for Ceta, can not be used in fast_mode

#-----output data folder setting-----
#name of folder for saving frames
folder_name @= 20250727_YtTomo_yq

# map output file path and
work_dir @= Z:\user_data\%folder_name

# file path for saving frames, not Falcon or DE camera, e.g. K2/K3 camera only
frame_dir @= Z:\user_data\%folder_name

```

Blue Box — Output Paths

Specify the output directory for data files.

Ensure that the map files and data output files are stored in separate directories to avoid overwriting or file management conflicts.

```

#-----data collection mode setting-----
data_mode = 1 #set 1 for dose sym tomography
               #2 for one or two direction tomography
               #3 Single particle tilt to fix angle for data collection
               #4 Single particle tilt pair
fast_mode = 1 # 1 for using fast mode, SET FAST MODE as well
               #2 for not
accurate_position = 1 #1 for aligning each point to reference in V, 2 for not
use_focus_forZ = 2 #set 1 for focus every point to measure Z
                 #2 for not
do_eucentric_fine = 1 #set 1 for do eucentric fine each group
                    #2 for not
align_low_angle_only = 1 #1 for only align at first 4 tilts
                        #2 for align at all tilts

skip_value_limit = -100 # Value of buffer mean count to skip, lower than this value will skip collection
                       #use "ReportMeanCounts" to check value

background_stage = 1 #1 for use file_checker for stage movement when imaging
                    #2 for not

move_stage_instead_IS = 2 # 1 for only moving stage, 2 for using image shift besides stage
                        # 3 for only using image shift at high mag

#-----multi shot per hole for SPA setting-----
in_hole_distance = 0 # distance away from the selected point in micrometer, only for SPA, not for tomography and tilt pair
in_hole_num = 1 # num of image taken around the selected point, only for SPA, not for tomography and tilt pair
in_hole_start_angle = 0 # angle for the first point away from selected point, the only for SPA, not for tomography and tilt pair

#-----FAST MODE setting-----
R_delay = 0.9 # exposure time for Record mode in FAST mode
blank_delay = 0.72 # delay time in sec for the beginning of continous record mode
ObjVslSmatrix = {-0.000005 0.000049 -0.000011 -0.000050} #do Focus/Tune -> "Calibrate Coma vs Image Shift" and cope the output ObjVslSmatrix in log

```

Red Box — CR-BIS Core Parameters

To enable CR-BIS mode, set: **fast_mode = 1**

- **R_delay** — Exposure time for each target point under CR-BIS mode.
- **blank_delay** — A microscope-dependent parameter that compensates for camera model-specific delays.
 - For K3 cameras: when switching to Continuous Mode, a short delay occurs that may cause the first target exposure to be skipped. This value should therefore be included as <blank_delay>.
 - For Falcon4i cameras: there is no switching delay. However, the beam is briefly turned on at the focus position before the first target exposure, requiring an additional buffering time in the total exposure time formula.
- **ObjVslSmatrix** — Calibration matrix between objective lens deflection and image shift, used for coma and image-shift correction.
- For CR-BIS-Tomo mode, set **align_low_angle_only = 1**. This ensures the first five tilt angles use standard acquisition to obtain better alignment parameters.

Yellow Box — Fine Alignment Parameters

These parameters improve positional accuracy and are particularly important for cryo-ET.

- **accurate_position** — Optional for SPA acquisition; recommended for cryo-ET.
- **use_focus_forZ** — Measures focus at each target point at the starting tilt angle to obtain Z-height.
 - Ensure the focus position differs from the actual imaging region.
 - Recommended for grid-based cryo-ET; not suitable for FIB-lamellae.
- **do_eucentric_fine** — Performs fine eucentric height adjustment.
 - Recommended for non-lamella cryo-ET samples.
 - Not recommended for lamella-based cryo-ET due to potential occlusion at high tilt angles.

Purple Box — Background Stage Movement

The `background_stage` parameter allows pre-movement of the sample stage while CR-BIS exposures are running. This utilizes the idle camera buffering period after a long CR-BIS exposure to move the stage to the next tilt angle or focus group, saving acquisition time. To enable this feature:

1. Copy **file_checker.py** to the directory where the map files are stored.
2. Before starting acquisition, run:

python3 file_checker.py

This script monitors the exposure progress and automatically triggers stage movement once the exposure group finishes.

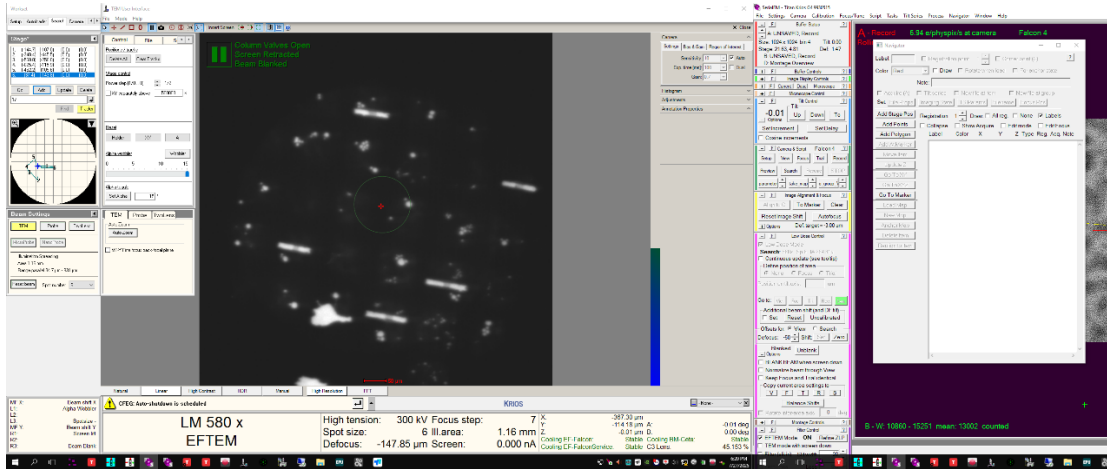
Initialization

After all parameters have been properly set, run the parameter script once. This will initialize the configuration and automatically create a data acquisition directory under the specified path for the current data collection session.

2.3 Low-Magnification Map Acquisition

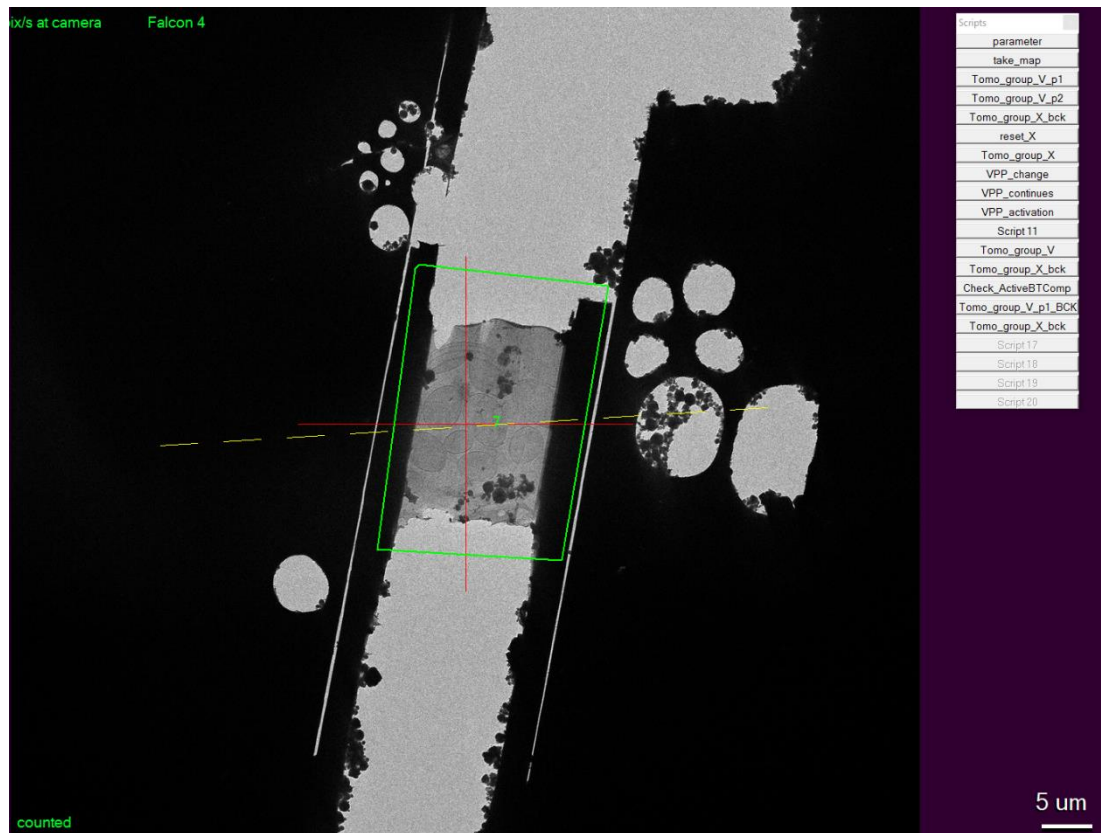
The procedure for acquiring low-magnification maps in **CR-BIS** is essentially identical to conventional cryo-EM data collection. The steps are as follows:

1. At low magnification, locate the target **mesh** or **lamella**.
2. Adjust the Z-height manually or automatically using **Tasks → Eucentric** to bring the specimen to the eucentric position.



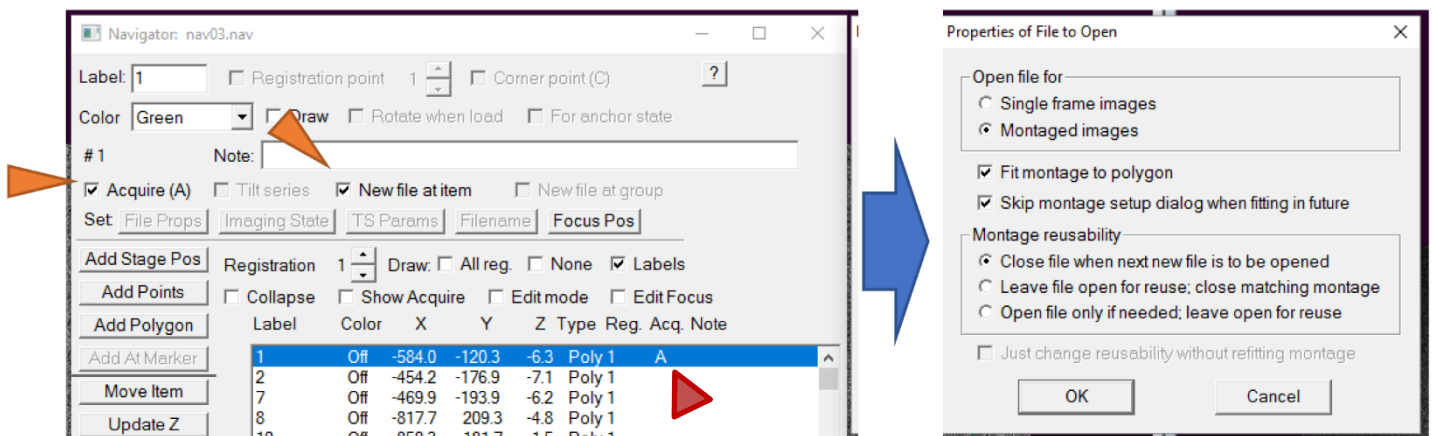
Low-magnification overview showing mesh or lamella locations. Green boxes indicate areas selected with Navigator → Add Polygon for subsequent high-magnification data collection.

3. In the **Navigator**, outline the desired acquisition regions using **Add Polygon**. Each polygon defines a region (e.g., one lamella or grid square) for subsequent targeting.



A representative lamella imaged at low magnification. The green polygon outlines the acquisition area; red or yellow annotations mark excluded or reference regions.

4. Switch to **View** mode, and set exposure parameters such as exposure time, binning, and gain reference.
5. Select the polygon(s) for mapping and check the “**Acquire**” (A) box. Specify an output filename (e.g., map01.mrc).



Left: start of a new target group, showing the first (focus) point and subsequent acquisition points. Right: end of the same group after all points are defined.

6. Execute **Navigator** → **Acquire at Items** to record the low-magnification map.

2.4 Target Selection

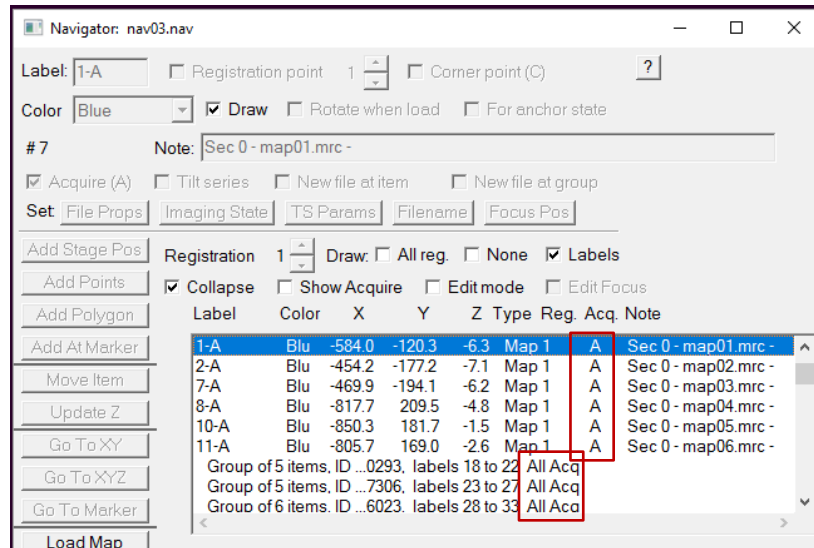
Once maps are acquired, define imaging targets in groups for automated **CR-BIS** collection.

1. Double-click the **map** in the Navigator to open it.
2. Select targets either manually or automatically, organizing them into **groups**. Each group should contain several nearby points.
3. The **first point of each group** serves as the *alignment/focus position* and is **not recorded** as data.
4. The remaining points in the group are acquisition targets. Recommended distances between the first and subsequent points: **Lamella (cryo-ET)**: < 6 μm ; **Single-particle (SPA)**: up to 10 μm (maximum practical distance) Distances should respect the beam-shift and coma-correction range.
5. Continue defining new groups until all target regions are covered.

2.5 Production Data Acquisition

After all maps and targets are defined, proceed with **CR-BIS data acquisition**:

1. Verify that all maps and points are **checked with “A”** in the Navigator. If polygons also have “A” marks, **remove them** (only points should remain checked).

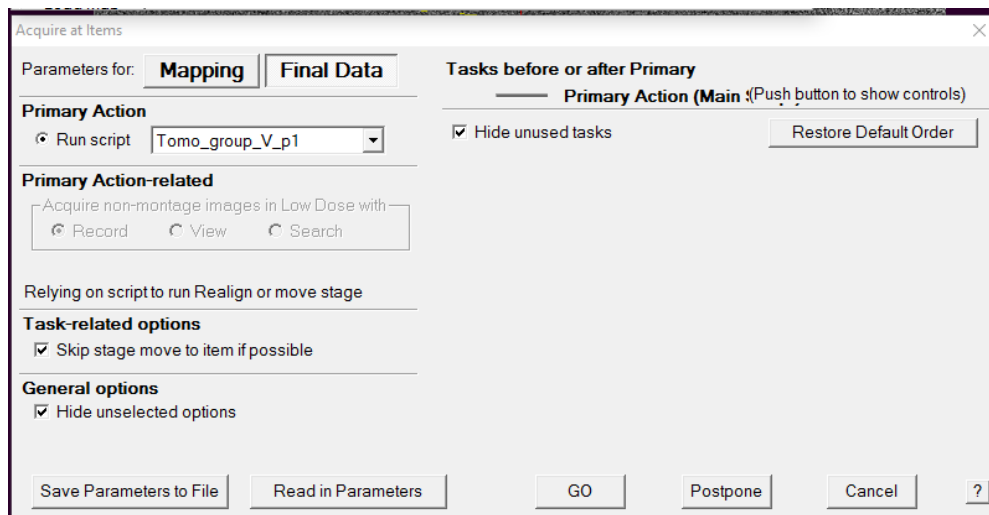


Ensure that each **Map item precedes its corresponding Points** in the Navigator list (maps must appear above their points).

2. Run the pre-alignment script via **Acquire at Items** →

FinalData(Tomo_group_V_p1) to initialize alignment and metadata recording.

- If acquisition is interrupted and the order of maps or points changes, **delete map_info.txt** before resuming to avoid indexing errors.



3. Start the main acquisition run using **Acquire at Items** →

FinalData(Tomo_group_X).

Acquire at Items

Parameters for: **Mapping** **Final Data**

Primary Action

Run script Tomo_group_X

Primary Action-related

Acquire non-montage images in Low Dose with

Record View Search

Relying on script to run Realign or move stage

Task-related options

☒ Skip stage move to item if possible

General options

☒ Hide unselected options

Tasks before or after Primary

Primary Action (Main) (Push button to show controls)

☒ Hide unused tasks Restore Default Order

Save Parameters to File Read in Parameters GO Postpone Cancel ?

4. Monitor the acquisition log and disk usage to confirm that frame splitting and data writing proceed normally.

Part 3: Preprocessing of CR-BIS Output Data

Two test datasets (CR-BIS-SPA and CR-BIS-Tomo) are provided for validation. All preprocessing scripts require Python 3.

3.1 Processing TIF-format Data

Script: group_single_point_tif_MotionCorr2_parallel_v2_tiff.py

Usage:

python3 group_single_point_tif_MotionCorr2_parallel_v2_tiff.py [options]

input_directory

Generates separated single-point files.

Required: input_directory (path to raw CR-BIS data).

3.2 Processing EER-format Data

Convert EER files to TIF (e.g., using *relion_convert_to_tiff_mpi*), then process as TIF files.

3.3 Processing Cryo-ET Data

For cryo-ET datasets, create image list or .mdoc files per tilt series. A reference workflow is provided in the CR-BIS-Tomo directory.